

# Educational Outcomes & Resource Allocation

Statistical Analysis Report

100-School Dataset · 5 Regions · 3 School Types

Generated May 2026

# Executive Summary

This report analyses a simulated dataset of 100 schools across five geographic regions and three school types (Public, Private, Charter). The goal is to identify which resource and staffing factors most strongly predict student academic performance. Key metrics examined include average test scores, graduation rates, pass rates, STEM scores, and literacy scores against resource variables such as budget per student, student-teacher ratio, teacher qualifications, computer access, and extracurricular programme offerings.

## Dataset Snapshot

|                                           |                                                 |                                                    |                                             |
|-------------------------------------------|-------------------------------------------------|----------------------------------------------------|---------------------------------------------|
| <div>85.1</div> <div>Avg Test Score</div> | <div>82.8%</div> <div>Avg Graduation Rate</div> | <div>\$5,136</div> <div>Avg Budget / Student</div> | <div>8.4%</div> <div>Avg Dropout Rate</div> |
|-------------------------------------------|-------------------------------------------------|----------------------------------------------------|---------------------------------------------|

## Summary Statistics

| Metric                    | Mean    | Min     | Max     | Std Dev |
|---------------------------|---------|---------|---------|---------|
| Avg Test Score            | 85.1    | 35.0    | 98.0    | 14.0    |
| Graduation Rate (%)       | 82.8    | 65.8    | 99.0    | 7.7     |
| Pass Rate (%)             | 75.5    | 30.0    | 92.4    | 13.7    |
| Dropout Rate (%)          | 8.4     | 0.5     | 14.7    | 3.8     |
| Budget / Student (USD)    | \$5,136 | \$1,663 | \$7,953 | \$1,835 |
| Student–Teacher Ratio     | 24.3    | 2.2     | 178.4   | 26.9    |
| Teacher Qualification (%) | 74.8    | 50.6    | 100.0   | 15.4    |
| Attendance Rate (%)       | 84.9    | 70.3    | 98.5    | 8.1     |
| Num Computers             | 256     | 28      | 482     | 136     |
| Extracurricular Programs  | 10.7    | 1       | 20      | 5.7     |

Table 1. Descriptive statistics across all 100 schools.

## Correlation Analysis

Pearson correlation coefficients were computed between resource/staffing variables and the primary outcome variable (Avg Test Score). Values range from  $-1$  (perfect negative) to  $+1$  (perfect positive). A threshold of  $|r| > 0.3$  is used to flag practically meaningful associations.

### Correlations with Average Test Score

| Variable                  | Correlation (r) | Strength    | Direction |
|---------------------------|-----------------|-------------|-----------|
| Student–Teacher Ratio     | $-0.737$        | Very Strong | Negative  |
| Budget per Student (USD)  | $+0.471$        | Moderate    | Positive  |
| Number of Teachers        | $+0.471$        | Moderate    | Positive  |
| Computer–Student Ratio    | $+0.260$        | Weak        | Positive  |
| Attendance Rate (%)       | $+0.200$        | Weak        | Positive  |
| Teacher Qualification (%) | $+0.159$        | Very Weak   | Positive  |
| Extracurricular Programs  | $+0.037$        | Negligible  | Positive  |
| Teacher Experience (Yrs)  | $+0.008$        | Negligible  | Positive  |
| Dropout Rate (%)          | $-0.031$        | Negligible  | Negative  |
| Enrollment                | $-0.379$        | Weak        | Negative  |

Table 2. Pearson correlation with Avg Test Score. Highlighted rows indicate  $|r| > 0.25$ .

### Key Findings

- **Student–Teacher Ratio ( $r = -0.74$ ):** The dominant predictor. Schools with high student loads per teacher consistently record the lowest scores. This is the most actionable metric for policymakers — reducing class sizes delivers the largest expected return.
- **Budget per Student ( $r = +0.47$ ):** A moderately strong positive link. However, budget alone does not guarantee outcomes; how funds are allocated (staffing vs. infrastructure) matters.
- **Enrollment ( $r = -0.38$ ):** Larger schools tend to score lower, likely as a proxy for higher student-teacher ratios in big institutions.
- **Attendance Rate ( $r = +0.20$ ):** Every percentage point improvement in attendance correlates with modest but consistent gains in test performance.
- **Teacher Qualifications ( $r = +0.16$ ):** Surprisingly weak. Credential percentage matters less than staffing quantity in this dataset — quantity of teachers outweighs formal credential level.
- **Extracurriculars & Teacher Experience ( $r \approx 0$ ):** Near-zero correlations suggest these variables, in isolation, do not reliably differentiate high from low performers.

## Budget Tier & Regional Analysis

### Performance by Budget Tier

Schools were segmented into four equal-width budget tiers to examine the dose-response relationship between spending and outcomes.

| Budget Tier              | N (approx) | Avg Test Score | Graduation Rate (%) | Pass Rate (%) |
|--------------------------|------------|----------------|---------------------|---------------|
| Low (\$1.5k–\$3.1k)      | 25         | 73.7           | 82.0                | 65.0          |
| Mid-Low (\$3.1k–\$4.7k)  | 25         | 82.7           | 82.4                | 73.7          |
| Mid-High (\$4.7k–\$6.3k) | 25         | 87.0           | 82.9                | 77.2          |
| High (\$6.3k–\$8.0k)     | 25         | 92.1           | 83.4                | 81.9          |

Table 3. Mean outcomes by budget tier. Test score improves monotonically with spending (+18.4 pts low→high).

### Regional Performance

| Region  | Avg Test Score | Graduation Rate (%) | Avg Budget/Student | Dropout Rate (%) |
|---------|----------------|---------------------|--------------------|------------------|
| South   | 89.5           | 84.8                | \$6,016            | 7.5              |
| West    | 88.2           | 83.3                | \$5,262            | 7.9              |
| North   | 86.3           | 80.4                | \$4,853            | 9.3              |
| Central | 82.4           | 84.8                | \$4,525            | 7.6              |
| East    | 79.4           | 80.8                | \$4,991            | 9.6              |

Table 4. Regional averages. South leads in test scores and budget; East trails in both.

### School Type Comparison

| School Type | Avg Test Score | Graduation Rate (%) | Pass Rate (%) | Budget/Student | Dropout (%) |
|-------------|----------------|---------------------|---------------|----------------|-------------|
| Private     | 86.7           | 82.3                | 79.5          | \$5,421        | 8.7         |
| Public      | 85.1           | 83.2                | 78.0          | \$5,014        | 8.4         |
| Charter     | 83.8           | 82.9                | 75.5          | \$4,992        | 8.3         |

Table 5. School type averages. Differences are modest; type is a weaker predictor than staffing or budget.

## Key Insights & Recommendations

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### 1. Student–Teacher Ratio Is the #1 Driver

A correlation of  $-0.74$  with test scores makes this the single most powerful signal in the dataset. Schools with ratios above 40:1 consistently fall below the 70-point mark. Every 10-student reduction in ratio is associated with approximately 7–9 extra score points. Policy recommendation: prioritise teacher hiring over infrastructure spending.

### 2. Budget Matters — But Allocation Is Everything

High-budget schools (\$6.3k–\$8k/student) average 92.1 vs. 73.7 for low-budget peers — an 18.4-point gap. However, budget correlates more weakly with graduation rate ( $r = +0.11$ ), suggesting that money directed at staffing yields stronger returns than capital expenditure.

### 3. South Region Leads; East Trails

The South averages the highest test score (89.5) and the highest budget (\$6,016), while the East has the lowest score (79.4) and the highest dropout rate (9.6%). Regional equity in funding distribution warrants attention from district administrators.

### 4. Qualification Rate Is a Weak Predictor

Teacher qualification percentage correlates only  $+0.16$  with scores. Schools with 100% qualified teachers do not consistently outperform those with 50%. This suggests that in-service training or staff count may be more impactful levers than credential gatekeeping.

### 5. Graduation and Dropout Are Near-Perfect Inverses

$r = -0.94$  between graduation and dropout rates is a near-perfect inverse. This is both a data quality indicator (the metrics are defined as complements) and a signal that retention strategies and graduation support likely address the same root causes.

### 6. Extracurriculars & Experience Show Negligible Links

Both extracurricular programme count ( $r = 0.04$ ) and teacher experience ( $r = 0.01$ ) show essentially zero correlation with test scores. This does not mean they lack value, but their effect may be mediated through other variables not captured in this dataset.

## Recommendations for Further Investigation

- Run **multiple linear regression** controlling for region and school type simultaneously to isolate net effects of each resource variable.
- Investigate **within-school longitudinal data** — does a budget increase at the same school lead to score improvement over time?
- Explore **non-linear thresholds** in student–teacher ratio (e.g., does the effect plateau above ratio 30:1?).
- Collect **socioeconomic context data** (median household income, free lunch programme %) to separate resource effects from socioeconomic confounders.
- Perform **cluster analysis** to identify school profiles (high-budget/high-ratio vs. low-budget/low-ratio) and target policy accordingly.